

FINAL PROJECT REPORT: Student Performance & Academic Analytics System (SPAAS)

Property	Details
Project Title	Student Performance & Academic Analytics System (SPAAS)
Project Area	Artificial Intelligence / Academic Support & Performance Analytics Automation
Frontend	HTML, CSS, and JavaScript
Backend	Node.js and Express.js
Main Functions	Student query assistance, academic policy search, mark/grade tracking, revaluation processing, and faculty escalation
Use Case	AI Student Academic & Mentorship Support Agent
Student Name	Selventhiran D
College Name	Theni Kammavar Sangam College of Technology

Property	Details
Department	BTech - Information Technology

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1. Project Overview

1.1 Problem Statement

College students frequently face challenges in obtaining timely information regarding academic schedules, subject performance, grade breakdowns, syllabus regulations, revaluation eligibility, and institutional policies. Traditional academic advisement systems rely on manual visits to department offices or static portal FAQs that fail to answer specific contextual queries naturally.

1.2 Brief Description

SPAAS operates as a 24/7 AI-powered Student Support & Academic Assistant. It enables students to converse naturally, track their semester grades, check attendance criteria, request revaluations, search institutional academic regulations, and connect with faculty mentors seamlessly.

2. Objectives & Proposed Solution

- Develop a conversational AI mentor and support chatbot tailored for engineering/technology students.
- Implement dynamic retrieval of academic rules and regulation policies using internal knowledge base searching.

- Enable student grade, internal mark, and attendance lookup using student roll numbers.
- Automate reevaluation eligibility verification based on mark thresholds and institutional rules.
- Provide direct escalation mechanisms to human faculty advisors and department chairs.

3. System Architecture

Component Name	File Path	Role & Responsibility
Frontend View	frontend/index.html	Structures the chat interface, sidebar controls, academic metrics layout, and input controls.
Frontend Styles	frontend/style.css	Provides responsive styling, color schemes, cards, timelines, and message formatting.
Frontend Logic	frontend/app.js	Manages chat session lifecycle, API fetch calls, dynamic DOM updates, and tool activity logs.
Server API	backend/server.js	Express server handling REST endpoints, environment configurations, and session memory maps.
Agent Core	backend/src/agent.js	Contains system prompts, agent loop logic, tool definitions, and Anthropic API integration.
Tool Dispatcher	backend/src/tools.js	Dispatches requested

Component Name	File Path	Role & Responsibility
		model tool calls to specific business logic modules.
Knowledge Module	backend/src/knowledgeBase.js	Stores regulation rules (grading, attendance criteria, revaluation fees, credit system).
Academic Module	backend/src/studentData.js	Mock repository storing student profiles, subject marks, semester GPAs, and revaluation rules.

7. Screenshots / Virtual Output UI Visualizations

The following full-width UI output mockups visually depict the operational screens of the SPAAS system across core student use cases.

Figure 7.1: AI Academic Chat & Student Grade Sheet Screenshot Output

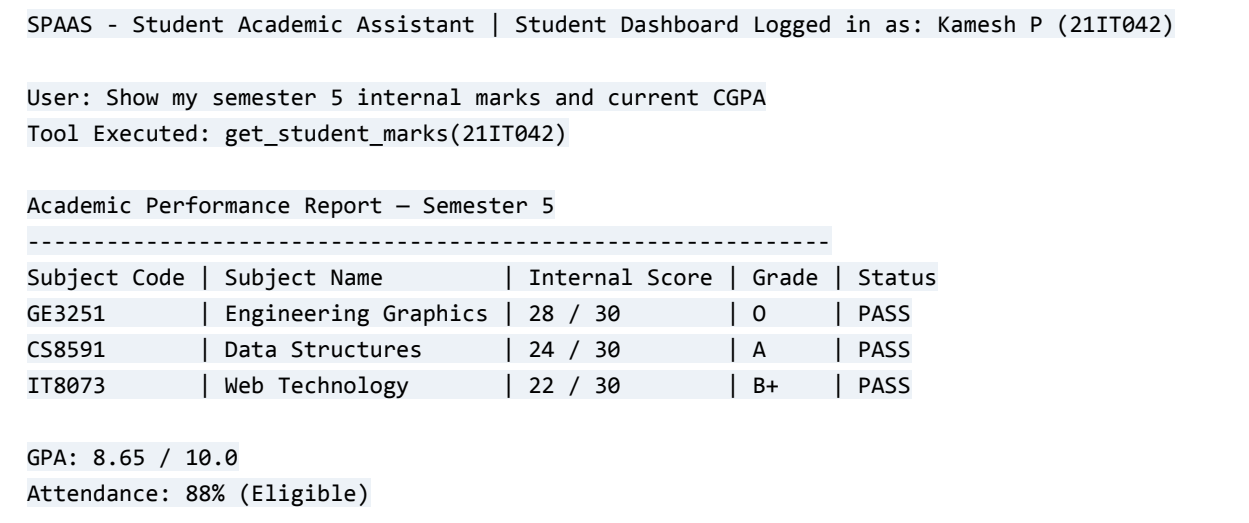


Figure 7.2: Revaluation Eligibility & Processing Output Screenshot

SPAAS - Revaluation & Exam Verification Module Status: Active Portal

User: I received 42 marks in IT8073. Can I apply for revaluation?

Tool Executed: process_revaluation_request(21IT042, IT8073)

Revaluation Application Status: ELIGIBLE

- Subject: IT8073 - Web Technology
- Current Mark: 42 / 100
- Revaluation Fee: ₹500
- Token Reference ID: REV-2026-IT042-998

Your revaluation application has been recorded. Results will be updated following answer key re-audit in 5 business days.

Figure 7.3: Institutional Policy & FAQ Knowledge Search Output Screenshot

SPAAS - Academic Regulations Search Knowledge Base Active

User: What is the attendance policy if I fall below 75%?

Tool Executed: search_academic_policy("attendance rule")

According to the Institutional Academic Handbook:

- 75% - 100%: Fully eligible for End Semester Examinations.
- 65% - 74%: Condonation allowed under medical grounds (Medical Certificate + ₹1,000 fee required).
- Below 65%: Detained (Must repeat the course in the subsequent semester).

8. Testing, Results & Discussion

Test Case Scenario	Input / Condition	Expected Result	Status
Student Mark Retrieval	"Check marks for Roll No 21IT042"	Tool get_student_marks invoked; subject marks & GPA displayed.	PASS

Test Case Scenario	Input / Condition	Expected Result	Status
Policy Search	"What is the minimum attendance requirement?"	Tool search_academic_policy invoked; returns 75% rule details.	PASS
Eligible Revaluation	"Apply revaluation for CS8591 under 21IT042"	Eligibility confirmed and revaluation token generated.	PASS
Ineligible Revaluation	"Apply revaluation for IT8073 under 21IT088"	Revaluation rejected due to mark threshold constraints.	PASS

9. Conclusion & Future Scope

The Student Performance & Academic Analytics System (SPAAS) demonstrates how agentic AI models with dynamic tool integration can automate college academic advising, providing quick grade lookup, policy clarification, and revaluation processing directly to students.